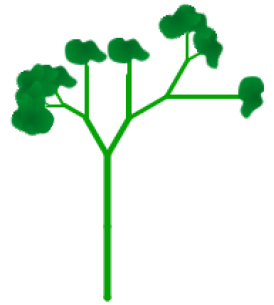


Scratchling

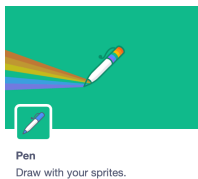


Register/login at <https://scratch.mit.edu>

Grow your own Scratch sapling – a scratchling.



1) Click on the **Add Extension** button, and add the **Pen** extension.



2) Create a sprite using the **Tree1** graphic.

3) In costumes, select one leaf and drag it to one side.

4) Delete the rest of the tree, then drag the leaf to the centre cross.



Make a sapling (a young tree).

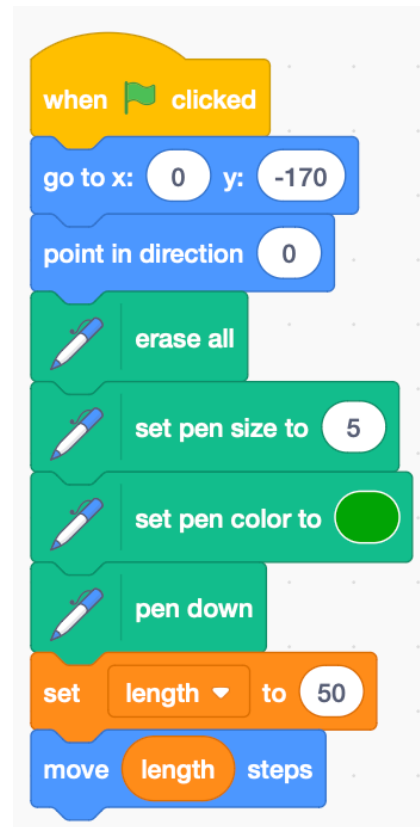
5) Make a variable, for the **length of the stalk**, for this sprite only.

6) Add the code to the right. First, place the sprite somewhere near the bottom of the screen, **pointing** upwards.

7) Use pen commands to **erase** the screen, set the pen **size** and **colour**, and put the **pen down** to start drawing.

8) Initialise the stalk **length** to 50, and then **move** it to draw the stalk.

Run this to see your new sapling.



Grow your plant with a repeat loop

9) **Make a block** to **grow** your plant.

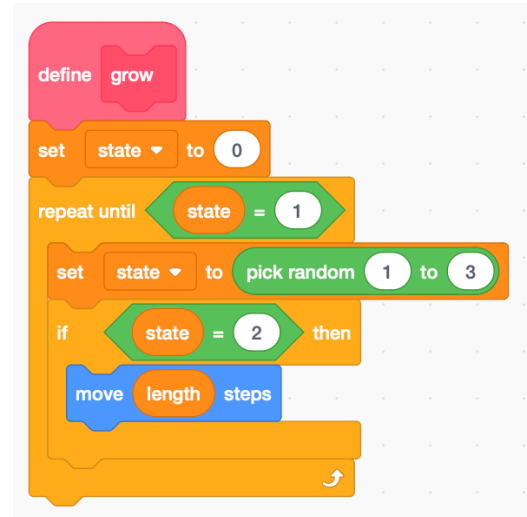
10) Make a variable for the **state** of the plant, **for this sprite only**, and initialise it to zero.

11) **Randomise** the **state** each time round the loop, between 1 and 3.

12) When **state** = 1 it exits the loop and stops growing.

13) **If state** = 2 **move** the sprite to grow a little bit.

14) Add  to the end of your **green flag** code.



Now the clever bit – the plant branches.

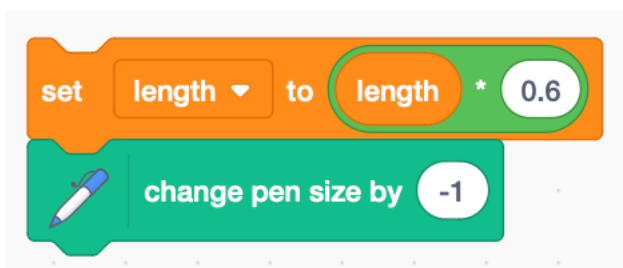
15) Add code after the last **if**, inside the loop, so the main plant **turns** 30° left and a **clone** of the sprite branches right.

16) Add code **when it starts as a clone** to **turn** 30° right, and continues to 



Is your plant a bit bushy? Prune it by cutting back the length each time round the loop.

17) Add code at the end of the loop – but inside it – to scale back the **length** and reduce the pen **size**.



Remember to **Save** your code with a good name.